

Strategy Name: Adaptive Macro Allocation Strategy

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Abstract

The Adaptive Macro Allocation Strategy is a diversified macroeconomic investment framework designed to balance capital preservation, income generation, and tactical exposure to real assets across changing market regimes.

The strategy combines investment-grade corporate bonds, U.S. Treasuries, gold, and commodity futures using a rules-based allocation system driven by macroeconomic indicators including U.S. Dollar strength (DXY), real yields, and Federal Reserve policy conditions.

Historical backtesting across major macroeconomic events, including the 2008 Global Financial Crisis, the COVID-19 market shock, and the 2022 inflation-driven tightening cycle, demonstrated that the strategy achieved a Compound Annual Growth Rate (CAGR) of 4.53% with annualized volatility of 8.36%.

Although the strategy underperformed the S&P 500 benchmark in total return, it exhibited lower volatility, improved diversification characteristics, and stronger downside resilience during periods of elevated macroeconomic uncertainty.

The findings suggest that adaptive macro allocation frameworks may provide investors with a lower-volatility alternative to traditional equity-heavy portfolios while maintaining long-term portfolio flexibility across changing economic environments.

Investment Thesis

Traditional long-term investment approaches often struggle to consistently outperform their benchmarks, while highly aggressive short-term trading strategies expose investors to excessive volatility and significant transactional costs.

The Adaptive Macro Allocation Strategy seeks to bridge this gap by combining defensive macro assets with selective tactical exposure to alternative markets. The strategy aims to generate attractive risk-adjusted returns through dynamic allocation across U.S. Treasuries, gold, and commodity futures markets.

The core premise is that macroeconomic regimes continuously shift between inflationary pressure, monetary tightening, economic slowdown, and risk-on expansion cycles. Different asset classes respond differently to these environments, creating opportunities for adaptive portfolio positioning.

The strategy focuses on:

- Capital preservation during market stress
- Opportunistic exposure to macro dislocations
- Low-to-moderate correlation with traditional equity benchmarks
- Tactical participation in commodity-driven trends

By combining defensive instruments such as U.S. Treasuries and gold with selective exposure to commodity futures, the strategy aims to achieve more stable long-term compounding while maintaining flexibility during rapidly changing market conditions.

Allocation Logic

The strategy utilizes a diversified macro allocation framework designed to balance capital preservation, income generation, and tactical exposure to real assets during changing economic conditions.

Target Allocation:

- 35% Investment-Grade Corporate Bonds
- 15% U.S. Treasuries
- 25% Gold
- 25% Commodity Futures

Portfolio Rationale

The fixed-income component, consisting of corporate bonds and U.S. Treasuries, provides portfolio stability, income generation, and partial downside protection during periods of economic uncertainty.

Gold serves as a defensive macro hedge during inflationary periods, geopolitical instability, and financial market stress. Historically, gold has demonstrated resilience during crisis-driven market environments and periods of declining confidence in risk assets.

Commodity futures provide tactical exposure to real assets that may benefit from inflationary cycles, supply-chain disruptions, and climate-related volatility. The inclusion of commodity exposure also aims to improve portfolio diversification through lower long-term correlation with traditional financial assets.

The strategy seeks to maintain a balance between defensive positioning and opportunistic macro exposure while reducing reliance on a single market regime or asset class.

Macro Signal Framework

The strategy utilizes a rules-based macroeconomic framework to dynamically adjust portfolio allocations according to changing monetary and market conditions.

- U.S. Dollar Regime (DXY)

The U.S. Dollar Index (DXY) is used as a primary indicator of global liquidity conditions and relative currency strength.

- Strong Dollar Environment:
Reduce commodity futures exposure by 5%
Increase defensive fixed-income allocation by 10%
Maintain moderate gold positioning (keep 25%)
- Weak Dollar Environment:
Increase commodity and gold exposure (increase both by 5%)
Reduce defensive bond positioning (decrease by 5%)

Historically, stronger U.S. Dollar environments have placed downward pressure on globally priced commodities, while weaker Dollar regimes have supported commodity appreciation and inflation-sensitive assets.

Real Yield Regime

Gold allocation decisions are primarily driven by movements in U.S. real yields.

Real Yield = Treasury Yield - Inflation Expectation

- Rising Real Yields:
- Reduce gold exposure (by 5%)
- Favor short-duration fixed-income instruments (T-Bills as was discussed earlier)
- Falling Real Yields:
- Increase gold allocation (by 2%)
- Increase duration exposure within Treasuries (sell 15% of T-Bills and buy 10-20 Years Treasury Notes)

Since gold does not generate cash flow or coupon income, declining real yields historically improve the relative attractiveness of gold as a store-of-value asset.

1. Federal Reserve Policy Regime

The strategy monitors Federal Reserve policy expectations using:

- CME FedWatch probabilities
- FOMC statements and meeting minutes
- Treasury yield curve behavior
- Hawkish Monetary Environment:
- Prioritize capital preservation
- Maintain shorter-duration Treasury exposure
- Reduce cyclical commodity allocation
- Dovish Monetary Environment:
- Increase duration exposure
- Expand commodity allocations selectively
- Increase gold positioning during declining real-yield environments

The objective of the framework is to systematically adapt portfolio exposure across multiple macroeconomic regimes while maintaining diversified downside protection and tactical flexibility.

Risk Management Framework

The strategy prioritizes capital preservation through diversified exposure limits and dynamic risk controls.

Key Risk Parameters:

- Maximum commodity futures exposure: 30%
- Minimum defensive fixed-income allocation: 40%
- Monthly portfolio volatility target: 8–12%

- Maximum target drawdown: 15%

Risk management procedures include:

- Monthly portfolio rebalancing last trading day only if allocation drift > 5%.
- DXY > 101 for 5 trading days
- Duration adjustments during periods of elevated rate volatility
- Exposure reduction during abnormal market stress conditions
- Liquidity prioritization through highly liquid ETFs and Treasury instruments

The strategy avoids concentrated exposure to any single commodity, issuer, or macroeconomic factor.

Signal Priority Matrix

1. Fed Policy
2. Real Yield
3. DXY

Precise execution thresholds

DXY above 101, strong dollar.

Rising Yield above 1.5% U.S. 10 Year Treasury

Commodity futures basket

Energy — 40%

- Brent (25%)
- WTI (West Texas Intermediate) (10%)
- Nat Gas (5%)

Metals — 30%

- Copper (5%)
- Precious metals (20%)
- Aluminum (5%)

Agriculture — 30%

- Coffee (15%)
- Wheat (10%)
- Corn (5%)

Benchmark: 60/40 Portfolio (60% stocks, 40% bonds) and the S&P 500 Index

Performance Objective

☐ **Target volatility: 8–12%**

☐ **Target Sharpe > 1**

The strategy did not fully achieve its target Sharpe ratio objective during the tested historical period.

☐ **Target drawdown < 10%**

Transaction Cost Assumptions

The backtesting framework incorporates realistic transaction cost assumptions to reduce performance overestimation.

Assumptions include:

- ETF expense ratios
- Futures rollover costs
- Bid-ask spread assumptions
- Estimated slippage during portfolio rebalancing
- Monthly transaction commissions

The strategy assumes moderate liquidity conditions and prioritizes highly liquid instruments to minimize execution friction.

Backtesting Methodology

The strategy will be evaluated using historical simulation across multiple macroeconomic environments, including inflationary periods, monetary tightening cycles, recessionary conditions, and high-volatility market regimes.

The backtesting process will include:

- Monthly portfolio rebalancing
- Historical macro signal analysis
- Rolling regime-based allocation adjustments
- Transaction cost assumptions
- Benchmark comparison against a traditional 60/40 portfolio

The strategy will be tested across:

- The 2008 Global Financial Crisis

- The COVID-19 market shock
- The 2022 inflation and rate-hike cycle
- Periods of low inflation and economic expansion

Performance evaluation metrics include:

- CAGR (Compound Annual Growth Rate)
- Sharpe Ratio
- Maximum Drawdown
- Volatility
- Sortino Ratio
- Correlation to benchmark portfolios

Macroeconomic Data Sources

- Federal Reserve Economic Data (FRED)
- CME FedWatch
- Treasury Yield Data
- DXY historical data
- Commodity futures market data

Instrument Universe

	Asset	Instrument
Gold		GLD
Treasuries		TLT / SGOV
Corp Bonds		LQD
Commodities		DBC

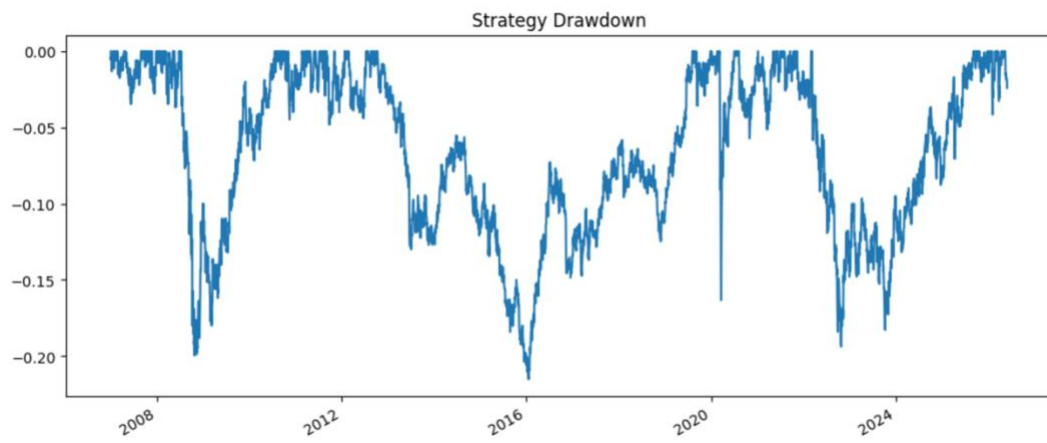
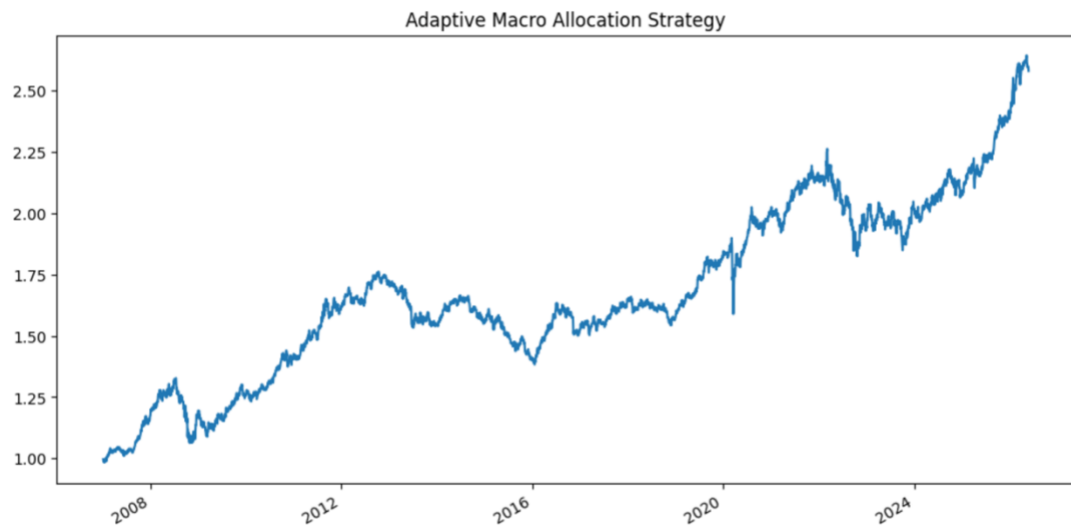
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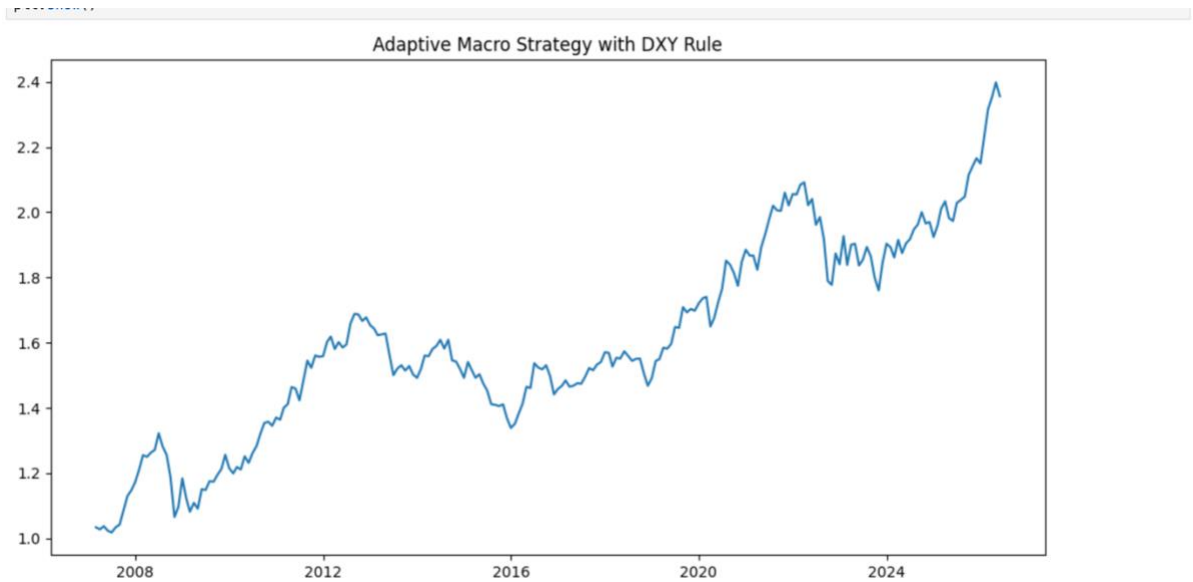
- Figure 1. Adaptive Macro Allocation Strategy Equity Curve
- Figure 2. Strategy Drawdown
- Figure 3. Adaptive Macro Strategy with DXY Rule
- Figure 4. Strategy vs Benchmark 60/40
- Figure 5. Rolling 12 months volatility
- Figure 6. Asset Correlation Matrix
- Figure 7. Strategy vs 60/40 & the S&P 500 Index benchmarks

Asset

Instrument

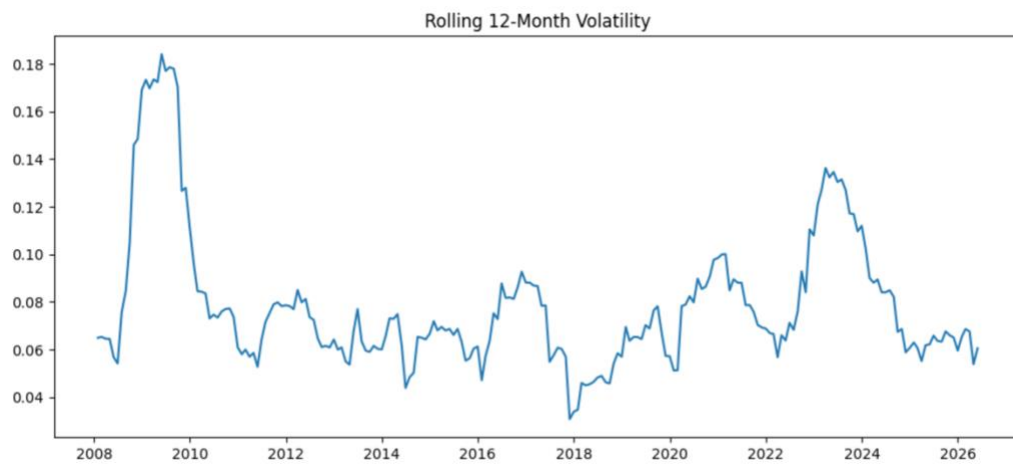
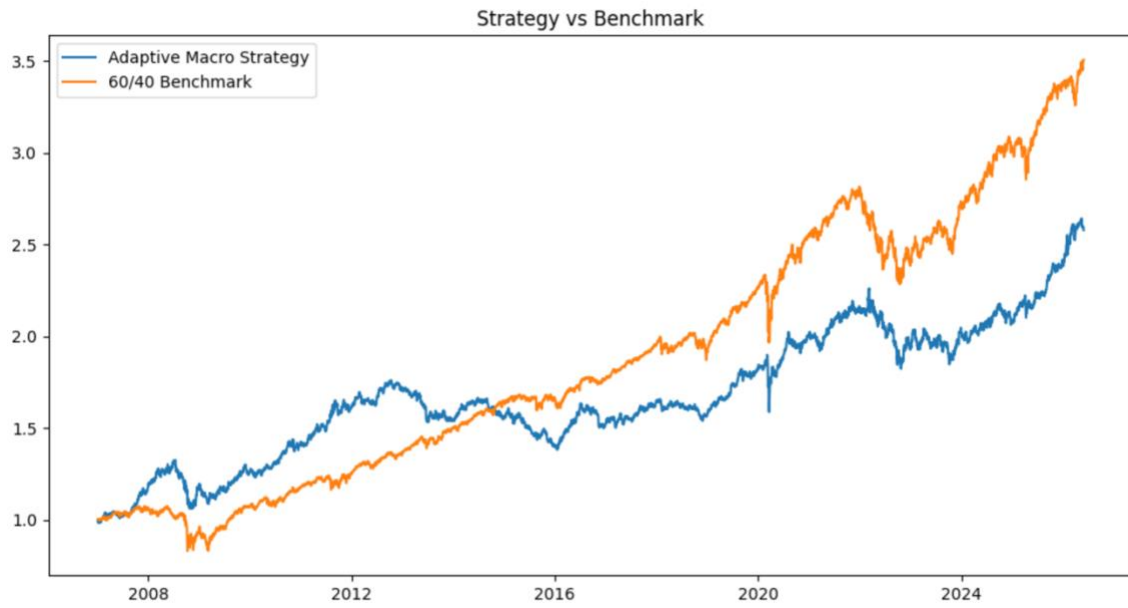
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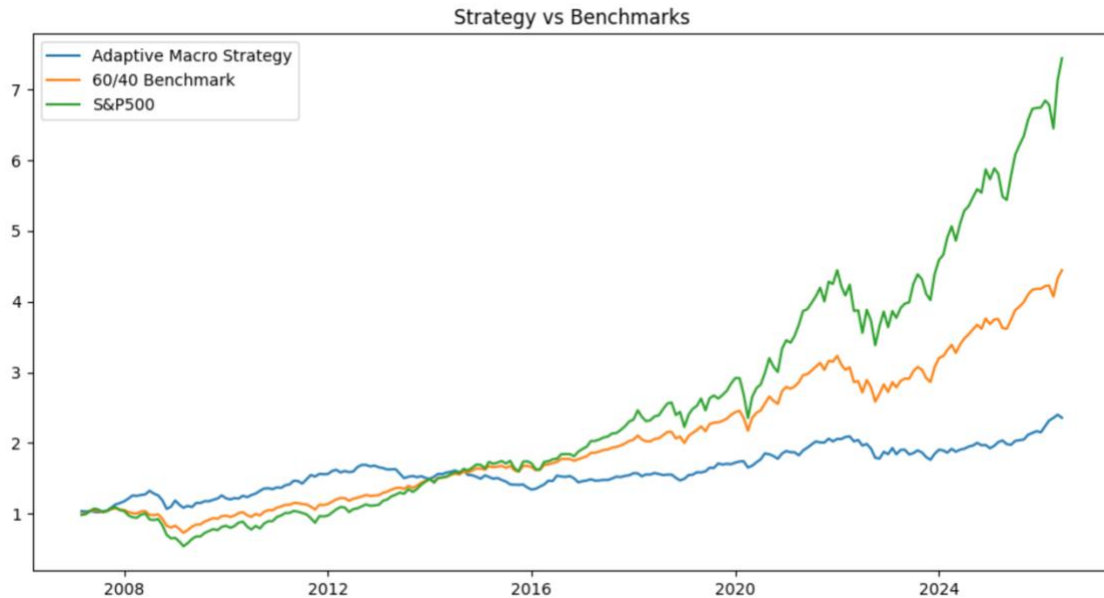


The adaptive macro allocation strategy achieved a CAGR of 4.53% with relatively low annualized volatility of 8.36%. The strategy's maximum drawdown remained limited to 20.71%, significantly lower than broad equity market benchmarks during crisis periods. This suggests that diversification across bonds, gold, and commodities improved downside protection while maintaining positive long-term returns.

Although the strategy did not fully achieve its target Sharpe ratio and drawdown objectives, the results demonstrate improved downside resilience and lower volatility relative to traditional equity-heavy portfolios during macroeconomic stress periods.







Limitations

- Historical relationships between macro variables may change over time.
- Backtesting does not guarantee future performance.
- Transaction costs and slippage may vary during stressed market conditions.
- The strategy relies on publicly available macroeconomic indicators that may lag real-time market behavior.

Conclusion

The Adaptive Macro Allocation Strategy demonstrated relatively stable long-term performance across multiple macroeconomic environments, including the 2008 Global Financial Crisis, the COVID-19 market shock, and the 2022 inflation-driven tightening cycle.

The strategy achieved a Compound Annual Growth Rate (CAGR) of 4.53% with annualized volatility of 8.36%. Although the strategy underperformed the S&P 500 benchmark in terms of total return, it exhibited significantly lower volatility and improved diversification characteristics. The results suggest that the strategy may provide more stable portfolio behavior during periods of elevated macroeconomic uncertainty.

The rolling volatility analysis showed that portfolio volatility remained relatively controlled during major market stress periods compared to traditional equity-heavy portfolios. In addition, the drawdown analysis demonstrated that diversification across Treasuries, corporate bonds, gold, and commodities contributed to downside risk mitigation during crisis environments.

The asset correlation matrix indicated relatively low correlations between most portfolio components, supporting the diversification benefits of the allocation framework. In particular, the negative relationship between defensive fixed-income instruments and commodity-related exposures helped reduce overall portfolio concentration risk.

Benchmark comparison against both the traditional 60/40 portfolio and the S&P 500 highlighted the trade-off between higher long-term equity returns and lower-risk macro diversification strategies. While the S&P 500 generated stronger overall growth, the Adaptive Macro Allocation Strategy maintained more stable risk-adjusted behavior and lower sensitivity to equity market shocks.

Overall, the findings suggest that macro regime-based allocation strategies may offer investors an alternative approach to traditional portfolio construction by emphasizing capital preservation, diversification, and adaptive positioning during changing monetary and economic conditions.

However, the strategy remains subject to several limitations, including historical data dependency, transaction costs, changing macroeconomic relationships, and potential execution constraints during stressed market environments. Future improvements may include more advanced signal modeling, dynamic volatility targeting, and broader global asset exposure.

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